

CV

Nicholas M. Nunley | nicknunley17@ucla.edu | [GitHub](#) | [Linkedin](#) | Los Angeles, CA

Technical founder, data scientist, software engineer, and molecular biologist passionate about precision medicine, information theory, mathematics, problem solving, and operating at the business/product interface

Experience

Founder & E-commerce Technical Consultant

Nunley Solutions (2026 - Present)

- Independent technical consultancy providing Shopify-native e-commerce engineering (custom Liquid and app development), automation, and analytics for client businesses
- Architected and led development and deployment of a dedicated wholesale Shopify storefront, including a custom Shopify app enforcing bespoke purchasing logic
- Designed and built a fully Shopify-native customer incentive program using Shopify Flow and Liquid
- Migrated ~7,200 flat B2B customer records into hierarchical, Shopify-native "company" data objects to support B2B commerce workflows
- Built operational automations spanning and surrounding Shopify, including a weekly one-way sync of ~95,000 customer records across multiple stores (Docker, AWS Fargate, AWS ECS, AWS Lambda) and GitHub Actions workflows for alerting
- Integrated and debugged third-party Shopify apps such as ShipStation for shipping and inventory management, Judge.me for reviews, BOGOS for promotional gifting, and Omnisend for marketing outreach
- Expanding into marketing analytics using GA4 to inform client marketing strategy
- Currently developing a product recommendation/calculator widget (FastAPI + Next.js + PostgreSQL) that maps user-input hair parameters to tailored product bundles

Research Data Scientist & Software Engineer (Staff Research Associate)

University of California Los Angeles (2022 - 2025)

- Hybrid data scientist, software engineer, and molecular/cell biology researcher leading computational biology, statistical analysis, and production-grade software development across two laboratories, spanning transcriptomic analysis and metabolic profiling, resulting in three peer-reviewed publications, two open-source packages, and a poster:
 - [Goldstein lab](#)
 - [Boutros lab](#)
- Owned and led two research projects in parallel and developed a third open-source software project spanning experimental design, data analysis, and software development
- Architected and released an R datasets package containing processed transcriptional data from 200+ samples across seven studies:
 - Performed data curation/QA (detecting and correcting sample swaps), built an object-oriented S4 dataset class, and added gene set enrichment and meta-analysis features plus a cross-species gene-identifier mapping system built without reliance on external databases
 - Engineered custom graphics solutions: resolved rendering failures via downsampling and coordinate offsets, extended lattice axes to support logarithmic scales, and designed a custom affine RGB colour-mapping engine
- Shipped RAGToolBox, a modular Python package for Retrieval-Augmented Generation (RAG) prototyping:
 - Engineered 9 modules and 4 CLI entrypoints, reaching ~83% test coverage, with 3 GitHub Actions CI/CD workflows automating build, test, and release
 - Released v0.1.0 with a tagged release and changelog, published to PyPI
- Mentored junior colleagues in Software Development Lifecycle (SDLC) practices and statistical modeling; conducted peer code reviews for data quality assurance, reproducibility, and analytical correctness
- Managed mammalian cell culture and functional assays (lentiviral transduction, immunoblotting, metabolic tracing, viability assays) supporting the above research

Lead K-12 Mathematics Instructor

Mathnasium, LLC (2019 - 2021)

- Provided one-on-one and group tutoring in foundational mathematics (e.g. algebra, geometry, trigonometry, calculus) and other general science courses
- Managed learning center operations on weekends:
 - Administered assessments to onboard new students
 - Managed student learning plans to track progress
 - Followed up with leads to convert enrollments
 - Scheduled and matched instructors with students to maintain coverage

Education

BS: Biochemistry

University of California Los Angeles (2020-2022)

- Conducted biomedical research while completing undergraduate coursework
- Completed a graduate-level course in mass spectrometry proteomics, and upper-division electives in mathematics and machine learning

AA (honors): Social and Behavioral Science

Santa Monica College (2015-2020)

- Explored multiple academic tracks before transitioning to STEM
- Worked part-time in service roles and as a peer tutor for foundational science and math courses

Skills

- Programming & scripting
 - Python programming
 - SQL
 - TypeScript
 - R programming
 - Unix
- Data science & analytics
 - Machine learning
 - Statistics
 - Optimization
 - Model evaluation & validation
 - Data visualization
 - Data modeling
 - AI Engineering
 - NumPy
 - pandas
 - Reproducibility
 - RAG
 - Network analysis
 - Simulation-based inference
 - Applied mathematics
- Software engineering & pipelines
 - git
 - CI/CD
 - Automation
 - API development
 - React web development
 - Next.js
 - FastAPI
 - PostgreSQL
 - Docker
 - AWS (Fargate, ECS, Lambda)
 - Testing (unit & integration testing)
 - Pipeline development and applications
 - High-performance computing
- Scientific computing & bioinformatics
 - Nextflow
 - Transcriptomics
 - Metabolomics
- E-commerce & business systems
 - Shopify Liquid
 - Shopify Admin API
 - Shopify Flow
 - Marketing analytics (GA4)
- Leadership
 - Project management
 - Problem formulation
 - Cross-functional collaboration
 - Mentorship
 - Scientific communication
- Experimental biology techniques
 - Biomedical experimental design

- Mammalian cell culture
- Immunoblotting
- Microscopy and IHC
- Cellular respirometry